

VKU Field Survey

Offline Data Collection PWA and Capacitor

A local-first campus facility inspection system for classrooms, laboratories, basements and remote buildings with weak or unavailable connectivity.

1. Problem And Objective

A network-dependent inspection form can lose unfinished observations, photos and submissions. This project treats IndexedDB as the primary source of truth: it autosaves drafts, accepts submissions offline, survives refresh/restart and synchronizes queued UUID records when the API is reachable.

The professional extension adds inspector/session accountability, category checklists, issue classification, versioned edits, GPS evidence, Admin review and Google Sheets reporting while preserving the original PWA and sync requirements. A server-validated Admin access key gates the review screen in both the PWA and APK.

2. Feature Checklist

DONE	Standalone manifest, VKU theme, 192/512 icons and app-shell precache
DONE	IndexedDB autosave, draft restore and schema-v1 compatibility
DONE	Local profile, active session and immutable survey snapshots
DONE	Six-step form, custom room, room type and 1-5 star rating
DONE	Five category-specific checklists with OK/Issue/Not Checked
DONE	Severity, priority, issue type, recommended action and evidence rules
DONE	Photo/GPS evidence with web and Capacitor implementations
DONE	Sequential offline queue, reconnect sync and idempotent UUID API
DONE	Filters, Needs Action badge, dashboard and local CSV/JSON export
DONE	Stable-UUID edits, version history and refresh-safe edit drafts
DONE	Admin assignment, review status, notes and central export
DONE	Server-validated Admin access key gate for PWA and APK
DONE	Google Sheets/Drive upsert, evidence URL and AuditLog
DONE	Java 21 APK build and public GitHub Release download
DONE	Public HTTPS PWA and same-origin Vercel API deployment
PENDING	Physical-device Camera, Network and GPS verification

Architecture And Offline Flow

3. Architecture

React UI (shared PWA / Android code)

- > IndexedDB v3: surveys | profile | sessions | surveyEdits
- > local-first submit: DRAFT -> PENDING_SYNC
- > foreground sync / Background Sync (sequential)
- > Express / Vercel REST API
- > Apps Script -> Google Sheets + Drive -> SYNCED

Client

React, TypeScript, Vite, Tailwind, React Router, `idb`, Workbox and Capacitor.

Server

Node.js, Express/Vercel API, Admin key middleware, version validation and an Apps Script central bridge.

4. Data Model

Each Survey includes UUID/version, sync and review statuses, inspector snapshot, session snapshot, facility location, room type, category checklist, rating, severity, priority, issue type, recommended action, notes, optional Base64 photo and GPS status/evidence.

Snapshots preserve historical accountability when the local profile or active session changes. IndexedDB version 3 also stores autosaved edits; a runtime normalizer safely reads original survey records.

5. Reliability Rules

- Draft changes are debounced and stored locally.
- Submission persists locally before any API request.
- Only a confirmed response changes status to Synced.
- Failures retain all data and remain retryable.
- A lock prevents overlapping foreground sync runs.
- Queued records upload in creation order; repeated UUIDs do not duplicate server data.
- Submitted edits retain UUID, increment version and return to Pending Sync.
- Admin review metadata stays separate from device synchronization state.
- Startup, browser/native network events and manual retry back up Background Sync.

6. PWA And Native Bridge

Workbox precaches HTML, CSS, JavaScript, icons and manifest data for offline boot. Browser evidence uses file input and Web Geolocation; Android uses Capacitor Camera, Network and Geolocation. GPS failure is recorded as unavailable and does not block a valid survey.

Java 21/Gradle builds package `edu.vku.fieldsurvey` for Android SDK 23-35. GitHub Actions signs and publishes the debug APK, so Android Studio is not required on the development machine.

From Field Entry To Resolution

The extension turns a simple form into a traceable inspection workflow. Every record answers four questions: who inspected it, where and when it was inspected, what condition was found, and what action should happen next.

7. Operational Modules

Module	Implemented behavior	Operational value
Inspector and session	Offline profile stores name, ID, class/unit, phone, role and team. An active session stores survey batch, date, campus/zone and shift. Both are copied as immutable snapshots into each survey.	Auditable ownership remains correct even after the next session or profile edit.
Facility context	Building, floor, room, room type and custom room values identify the inspected location without forcing a fixed room list.	Suitable for classrooms, labs, offices, halls, libraries and remote buildings.
Category checklist	Hardware, Projector, AC, Electrical and Furniture each load a focused checklist. Items support <code>OK</code> , <code>Issue</code> and <code>Not Checked</code> , while the original 1-5 rating remains available.	Inspectors record repeatable evidence instead of relying only on free text.
Issue management	Severity, priority, issue type and recommended action classify findings. High/Critical records require explanatory notes and a photo before submission.	Creates a clear hand-off from observation to repair, replacement, cleaning, escalation or monitoring.
Evidence and correction	Photos are resized and saved offline; GPS stores coordinates, accuracy, capture time and an unavailable status when the device cannot provide a fix. Submitted surveys can be edited with a stable UUID and incremented version.	Preserves evidence in weak-signal areas and keeps corrections linked to the original finding.
Review and reporting	History filters by status, category, severity, priority, inspector and date. Admin can assign, add review notes and move a survey through Open, In Review, Resolved or Rejected.	Turns collected data into an actionable work queue and a reportable dataset.

8. Status Lifecycle And Access

DRAFT -> PENDING_SYNC -> SYNCED -> IN_REVIEW -> RESOLVED / REJECTED
 \-> SYNC_FAILED -> RETRY

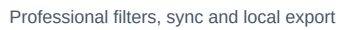
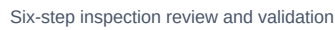
Inspector workflow

Complete profile/session, create a survey, move through the six steps, review validation, save locally, submit and continue working offline. History exposes retry, edit and export actions.

Admin workflow

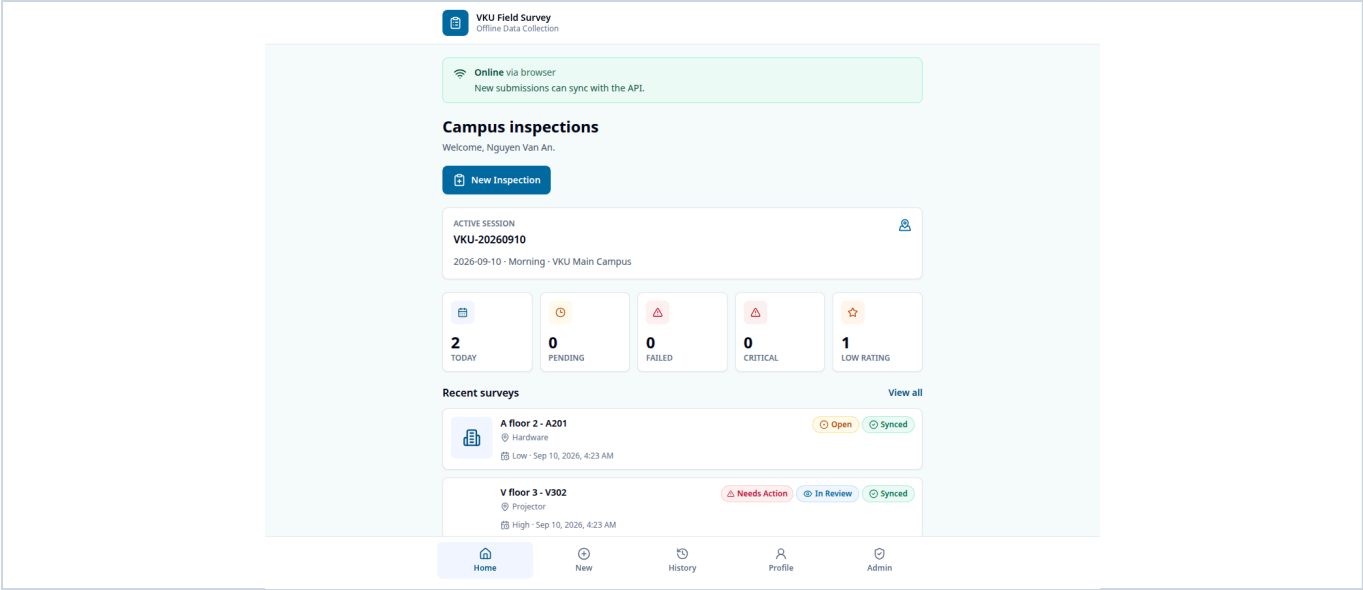
Open `/admin` , enter the demo key, load synchronized surveys, inspect evidence and update assignment/review metadata. Admin endpoints reject missing or invalid keys on the server.

Boundary for this mini-project: the Admin key is a controlled demo gate, not a replacement for production authentication or role-based identity management. It is configured by `ADMIN_ACCESS_KEY` and works in both the PWA and APK builds.



Screenshots are generated by the repeatable Chromium E2E workflow at a 390 x 844 mobile viewport.

Results



10. Automated Verification

Unit / Integration Business rules, High/Critical photo requirement, old-schema normalization, IndexedDB edit recovery, immutable snapshots, queue ordering and server/review validation.	Browser E2E Profile refresh, edit versioning, Admin review, online/offline submit, reconnect sync, GPS, export, wrong-key rejection and offline boot.
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Check	Result
<code>npm test</code>	Server and client unit suites pass, including validation, migration, queue, snapshot, edit and review rules.
<code>npm run check</code>	TypeScript and project checks pass for client, server and shared configuration.
<code>npm run test:e2e -w client</code>	Chromium flow passes for offline boot, reconnect sync, profile recovery, edit versioning, export, GPS and Admin key rejection/acceptance.
APK build	Capacitor Android build produces package <code>edu.vku.fieldsurvey</code> ; physical Camera/Network/GPS verification remains the final manual step.

11. Delivery Links

Deliverable	Access
Live PWA	https://miniproject1-client.vercel.app/
Admin screen	https://miniproject1-client.vercel.app/admin Demo key: VKU - ADMIN - 2026
GitHub source	https://github.com/tuaw-khoi/miniproject1
Android APK	https://github.com/tuaw-khoi/miniproject1/releases/download/apk-latest/vku-field-survey.apk
Google Sheet	https://docs.google.com/spreadsheets/d/1XcteJ_SH5Uo99ND0IzJh8EFS4Yfx-EnvC1Y8i9deh0Q/edit?usp=sharing
Drive evidence	https://drive.google.com/drive/folders/1zlXHJdZi2g7GdeCwiGG_xjk14bGyTDJ5?usp=sharing

12. Delivery Status

The PWA, same-origin API, public source and Android release are ready for demonstration. Central records use Google Sheets upsert semantics and uploaded photos expose Drive URLs in the sheet. The remaining validation item is installing the APK on a physical Android phone and exercising native Camera, Network and GPS permissions end to end.

Conclusion. VKU Field Survey meets the original offline-first learning objectives and now models a credible inspection-to-resolution process. Its core guarantee remains unchanged: valid field data is saved locally first and never depends on immediate connectivity.